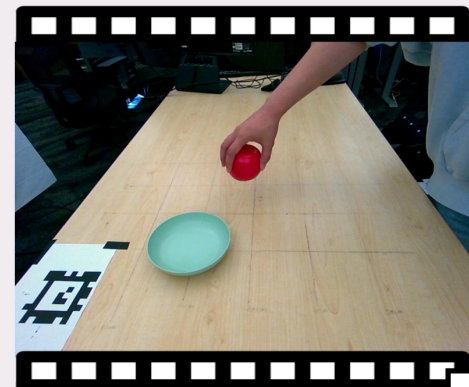


REAL-TO-SIM Off-the-Shelf Estimation



Single Human
RGB-D Video

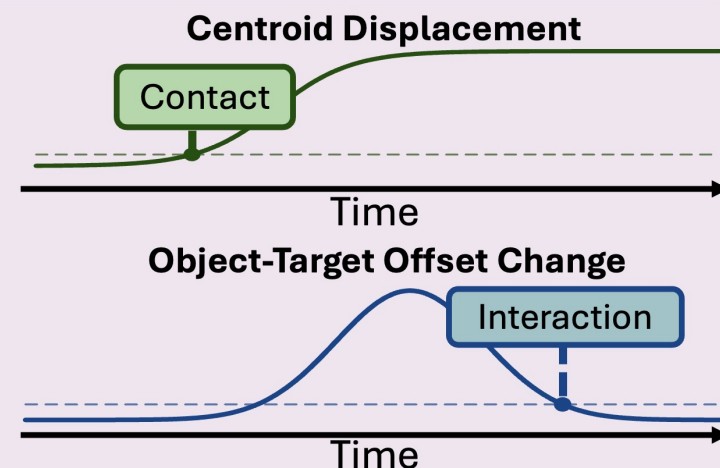
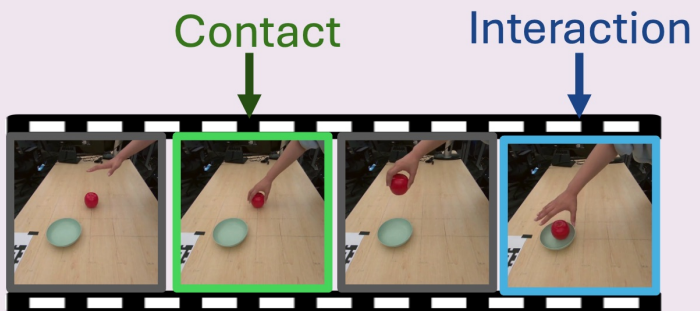
OUTPUT



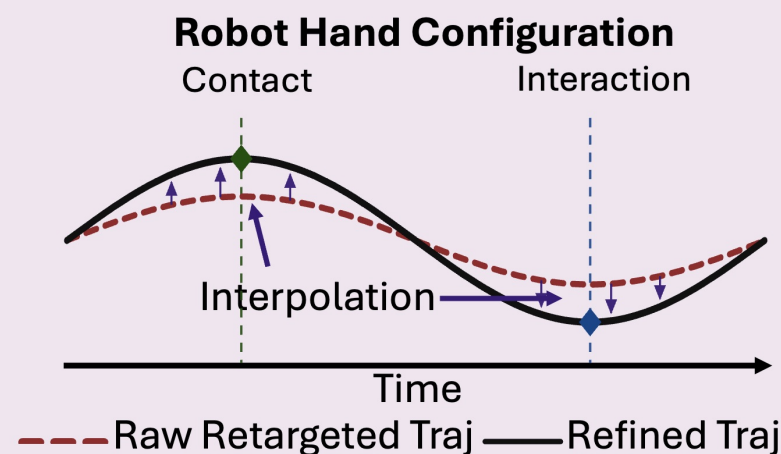
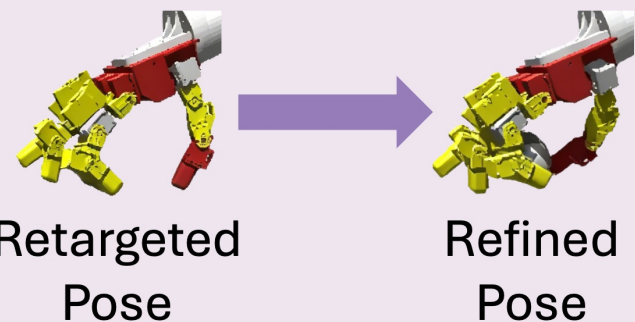
1

DIGITAL TWIN Keyframe Refinement

Keyframe Identification



Trajectory Refinement



Refined
Trajectory

OPTIONAL ADD-ON

Collision-Aware Motion Planner

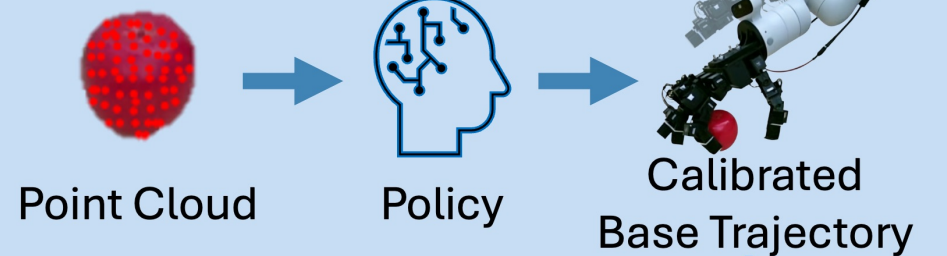
In: refined trajectory + obstacle scene + new task config
Out: collision-free trajectory

Collision-Free
Trajectory

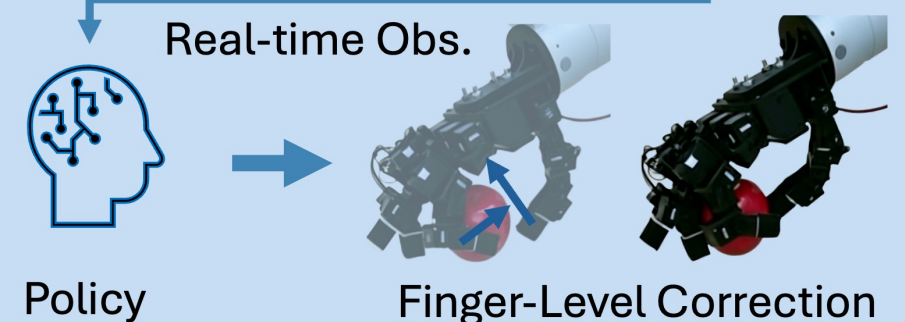
2

SIM-TO-REAL DEPLOYMENT Factorized Sim-to-Real Transfer

Global Geometry Adaptation



Local Contact Adaptation

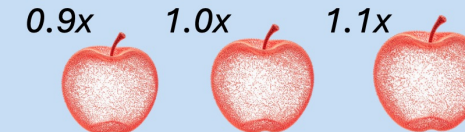


Object Pose

Point Cloud Noise

Domain
Randomization

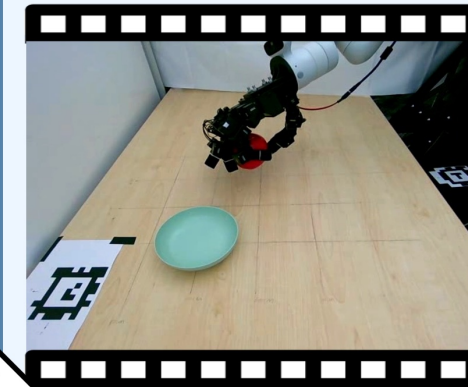
Object Scale



Object Dynamics



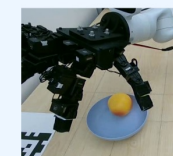
REAL ROBOT Dexterous Manipulation



Real-World Task
Execution

OUTPUT

Fruit
Preparing



Steak
Seasoning



Toy
Arrangement

